# Branch-Price-and-Cut: Origins, Theory, Components and Practice

With a research agenda for the Capacitated Profitable Tour Problem with Time Windows and Resource (Time) Constraints

Prepared for Afonso Fernandes

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**Abstract**

Branch-price-and-cut (BPC) combines three ideas: implicit enumeration by branch-and-bound (B&B), strengthening of linear relaxations by cutting planes, and solution of very large linear programs by column generation. This monograph traces how these ideas arose, states the theory that makes their combination valid, and describes the components that make modern BPC algorithms for vehicle routing work: restricted master problems with Farkas and reduced-cost pricing, dual stabilization, labeling algorithms for the elementary shortest path problem with resource constraints (ESPPRC), relaxations such as ng-routes, $q$-routes and decremental state-space relaxation, robust and non-robust cutting planes (rounded capacity, multistar, lifted cover, subset-row and limited-memory rank-1), Lagrangian bounds and reduced-cost fixing, route enumeration and branching. It surveys applications beyond routing and lists the parameters a BPC implementation must expose. The last part compares these components with an existing BPC code base and manuscript for the capacitated profitable tour problem with resource constraints and time windows (CPTP-RCTW) and proposes research directions, each marked as established transfer, moderate extension or speculative. All numbers quoted from the literature are taken from the cited sources; no computational results are invented.

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# 1 Introduction and reading guide

**Scope.** The term *branch-price-and-cut* (equivalently *branch-cut-and-price*, BCP) denotes an LP-based B&B in which the linear relaxation solved at each node is a Dantzig–Wolfe master problem with exponentially many columns, solved by column generation, and strengthened by valid inequalities separated during the solve (Desrosiers and Lübbecke, 2011; Costa et al., 2019). The method dominates exact vehicle routing: the generic solver VRPSolver handles a large family of routing and related problems in one framework (Pessoa et al., 2020), and the survey of Costa et al. (2019) organises the vehicle routing literature on the method.

**Structure.** Section 2 gives the lineage from B&B through branch-and-cut (B&C) and branch-and-price (B&P) to BPC. Section 3 formalises B&B. Section 4 collects the relaxations (LP, Lagrangian, Dantzig–Wolfe, state-space). Section 5 treats column generation in depth, including Farkas pricing and stabilisation. Section 6 covers pricing algorithms for routing. Section 7 covers cutting planes. Section 8 covers branching and primal heuristics. Section 9 assembles the full algorithm, Section 10 lists parameters and software, Section 11 surveys applications, and Section 12 gives the research agenda for the CPTP-RCTW.

**Conventions.** Problems are stated as minimisation unless they concern profits, where the CPTP-RCTW is written as maximisation to match the existing manuscript. $V$ denotes vertices, $A$ arcs, $E$ edges, $\delta(S)$ the edges with exactly one end in $S$, $\delta^+(S)/\delta^-(S)$ outgoing/incoming arcs, and $x(F) = \sum_{e \in F} x_e$. In Section 12, the tags [E], [M] and [S] mark an idea as an established technique to transfer, a moderate extension of published work, or a speculative proposal without published support.

# 2 History and lineage

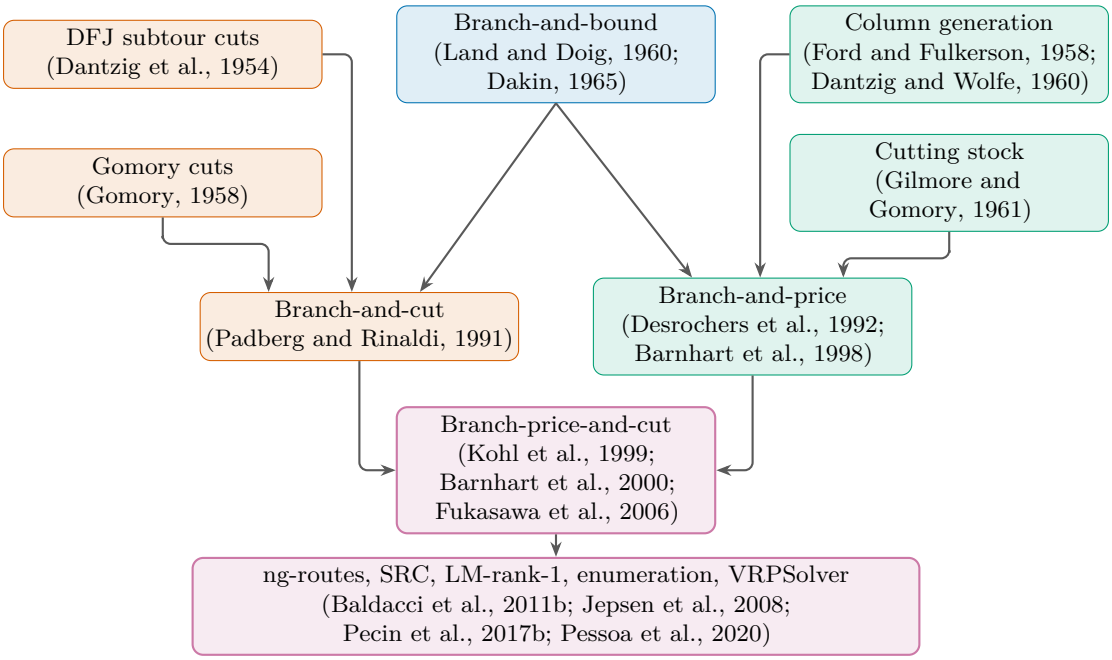



Figure 1: Lineage of BPC. Blue: enumeration; vermilion: cutting planes; green: column generation; purple: their combination. Arrows indicate conceptual descent, not citation counts.

**Branch-and-bound.** Land and Doig (1960) proposed an automatic method for discrete programming that solves LP relaxations, partitions the feasible region by fixing a fractional variable to integer values, and discards subproblems whose LP value cannot beat the best known integer solution. Dakin (1965) replaced Land and Doig's multi-way partition with the dichotomy $x_j \le \lfloor x_j^* \rfloor$ *or* $x_j \ge \lceil x_j^* \rceil$, the binary branching that LP-based solvers still use. The name "branch and bound" was used by Little et al. (1963) for their algorithm for the travelling salesman problem (TSP), whose bounds come from the assignment relaxation rather than an LP. Morrison et al. (2016) survey later advances in search, branching and pruning.

**Cutting planes.** Dantzig et al. (1954) solved a large TSP by adding subtour elimination constraints to an LP only when violated, an early instance of what is now called *lazy* constraint generation. Gomory (1958) gave a finitely convergent cutting-plane algorithm for pure integer programs; Kelley (1960) gave the analogous method for convex programs. Chvátal (1973) defined the closure and rank of the Chvátal–Gomory procedure, and Grötschel et al. (1981) proved that optimisation and separation over a polyhedron are polynomially equivalent, the theoretical basis for separating exponential families. Polyhedral work on knapsacks produced cover and lifted cover inequalities (Balas, 1975; Wolsey, 1975; Padberg, 1975), used in general 0-1 programming by Crowder et al. (1983).

**Branch-and-cut.** Combining cut generation with branching, Padberg and Rinaldi (1987) solved a 532-city symmetric TSP to optimality, and Padberg and Rinaldi (1991) described the resulting framework, for which they introduced the name branch-and-cut. Key features they emphasised—global cut pools, cuts valid throughout the tree, separation at every node, and reduced-cost fixing—remain the core of modern solvers such as SCIP (Achterberg, 2009).

**Column generation and Dantzig–Wolfe decomposition.** Ford and Fulkerson (1958) suggested generating path variables of a multicommodity flow LP on demand. Dantzig and Wolfe (1960) gave the decomposition principle: a linear program with a block-angular structure can be rewritten over the extreme points of the blocks, and the master LP solved by pricing out columns from subproblems. Gilmore and Gomory (1961, 1963) applied the idea to cutting stock, where the pricing problem is a knapsack. These works solved LPs; integer solutions were obtained by rounding.

**Branch-and-price.** Embedding column generation in B&B requires that the pricing problem remain tractable after branching. Desrochers and Soumis (1989) and Desrochers et al. (1992) solved crew scheduling and the vehicle routing problem with time windows (VRPTW) by column generation with shortest path pricing, and Desrosiers et al. (1995) surveyed time-constrained routing and scheduling by this approach. Ryan and Foster (1981) had already given the branching rule for set-partitioning masters that keeps pricing structure. Barnhart et al. (1998) unified the field under the name branch-and-price, discussing formulations, branching compatible with pricing, and the relation to Lagrangian relaxation; Vanderbeck and Wolsey (1996) and Vanderbeck (2000) gave general branching schemes for integer column generation. Monographs and tutorials include Desaulniers et al. (2005), Lübbecke and Desrosiers (2005) and Desrosiers and Lübbecke (2011).

**Branch-price-and-cut.** Cuts were added to B&P masters early: Kohl et al. (1999) added 2-path cuts to a VRPTW column-generation algorithm, and Barnhart et al. (2000) combined branch-and-price with lifted cover inequalities for integer multicommodity flow. Fukasawa et al. (2006) introduced *robust* branch-and-cut-and-price for the capacitated VRP (CVRP), where cuts are expressed in arc variables so that their duals only modify arc costs; the same principle was applied to the capacitated minimum spanning tree by Uchoa et al. (2008). Baldacci et al. (2008)

**Algorithm 1** LP-based branch-and-bound

$\mathcal{L} \leftarrow \{X\}$, $\bar{z} \leftarrow +\infty$, $\hat{x} \leftarrow$ none
**while** $\mathcal{L} \neq \emptyset$ **and** $\mathrm{gap}(\bar{z}, \min_k \underline{z}_k) > \varepsilon$ **do**
    select and remove $X_k$ from $\mathcal{L}$ ▷ node selection
    solve relaxation of $X_k$; obtain $\underline{z}_k$, $x^k$ ▷ bounding
    **if** infeasible **or** $\underline{z}_k \geq \bar{z}$ **then continue** ▷ prune
    **if** $x^k \in X$ **then** $\bar{z} \leftarrow \underline{z}_k$, $\hat{x} \leftarrow x^k$; **continue** ▷ prune by optimality
    run primal heuristics; update $\bar{z}, \hat{x}$ if improved
    choose a disjunction violated by $x^k$; add children $X_k^1, \ldots, X_k^m$ to $\mathcal{L}$ with bound $\underline{z}_k$ ▷ branching
**return** $\hat{x}$

used a set-partitioning formulation with additional cuts and route enumeration; Jepsen et al. (2008) introduced subset-row inequalities (SRIs), the first widely used *non-robust* cuts; Baldacci et al. (2011b) introduced ng-route relaxation. Pecin et al. (2017b) combined limited-memory rank-1 cuts, enumeration and other techniques, and Pecin et al. (2017a) extended them to the VRPTW. Pessoa et al. (2020) report in their introduction that these advances raised the size of CVRP instances solved from about 150 to 360 customers, and generalised the techniques into VRPSolver. The reference book on vehicle routing is Toth and Vigo (2014). Surveys include Baldacci et al. (2010), Baldacci et al. (2012), Poggi and Uchoa (2014), Desaulniers et al. (2014) and Costa et al. (2019).

# 3 Branch-and-bound

**Setting.** Consider $z^* = \min\{c^\top x : x \in X\}$ with $X = \{x \in \mathbb{Z}_+^n \times \mathbb{R}_+^p : Ax \geq b\}$. A *relaxation* is a problem $\underline{z} = \min\{f(x) : x \in Y\}$ with $X \subseteq Y$ and $f(x) \leq c^\top x$ on $X$, so $\underline{z} \leq z^*$. B&B maintains a list $\mathcal{L}$ of open subproblems $X_k \subseteq X$ whose union contains every optimal solution not yet found, an *incumbent* $\hat{x}$ with value $\bar{z} = c^\top \hat{x}$ (an upper bound), and the global lower bound $\underline{z} = \min_{k \in \mathcal{L}} \underline{z}_k$.

**The four operations.** *Bounding* computes $\underline{z}_k$ by solving a relaxation of $X_k$. *Pruning* removes $X_k$ if (i) the relaxation is infeasible, (ii) $\underline{z}_k \geq \bar{z}$ (*by bound*; with integer objective coefficients, $\lceil \underline{z}_k \rceil \geq \bar{z}$), or (iii) the relaxation optimum is feasible for $X$, in which case it updates the incumbent (*by optimality*). *Branching* partitions $X_k$ into $X_k^1, \ldots, X_k^m$ that exclude the current relaxation optimum. *Node selection* chooses which open node to process next. The algorithm stops when $\mathcal{L} = \emptyset$ or when the relative gap $(\bar{z} - \underline{z})/|\bar{z}|$ falls below a tolerance.

**Node selection.** *Best-first* (smallest $\underline{z}_k$) minimises the number of nodes processed given fixed bounds, because every node with $\underline{z}_k < z^*$ must be processed by any B&B using the same bounds. *Depth-first* finds incumbents quickly and reuses the parent's LP basis, which matters when re-optimisation is expensive. Hybrids dive depth-first and backtrack to the best node periodically (Achterberg, 2009; Morrison et al., 2016). In BPC, nodes are expensive and the initial upper bound is often supplied by a heuristic, so best-first search is the usual choice in VRP codes.

**Variable selection.** *Most-fractional* branching is cheap but on standard benchmarks no better than random (Achterberg et al., 2005). *Strong branching* solves the child LPs of several candidates and picks the one maximising a score of the two bound improvements, often the product $\max\{\Delta^-, \epsilon\} \cdot \max\{\Delta^+, \epsilon\}$. *Pseudo-cost* branching (Benichou et al., 1971) estimates $\Delta^\pm$ from historical per-unit bound changes; *reliability* branching (Achterberg et al., 2005) uses strong branching until a candidate's pseudo-cost has been observed a set number of times.

**Why bounds matter.** The size of the tree is governed by the gap between the root bound and $z^*$ and by how fast branching closes it. BPC invests heavily in bounding—an exponentially large formulation plus cuts—so that trees stay small even though each node is costly.

# 4 Relaxations

**LP relaxation.** Dropping integrality gives $z_{LP} = \min\{c^\top x : Ax \geq b, x \geq 0\}$. Its strength depends on the formulation; cutting planes tighten it towards the convex hull of $X$.

**Lagrangian relaxation.** Split constraints into "complicating" $Ax \geq b$ and "easy" $x \in X' = \{x \in \mathbb{Z}^n_+ : Dx \geq d\}$. For multipliers $u \geq 0$,

$$L(u) = \min_{x \in X'} \; c^\top x + u^\top (b - Ax) \leq z^*, \tag{1}$$

and the Lagrangian dual is $z_{LD} = \max_{u \geq 0} L(u)$. $L$ is concave and piecewise linear. Held and Karp (1970, 1971) used it with 1-trees for the TSP; Held et al. (1974) validated subgradient optimisation, with the update

$$u^{t+1} = \max\{0,\; u^t + \theta_t (b - Ax^t)\}, \qquad \theta_t = \alpha_t \frac{\bar{z} - L(u^t)}{\|b - Ax^t\|^2},\; \alpha_t \in (0, 2], \tag{2}$$

where $x^t$ solves (1) at $u^t$ and the step rule follows Polyak (1969). Fisher (1981) gives a classic tutorial; textbook treatments of relaxations and decomposition are Nemhauser and Wolsey (1988) and Wolsey (2020).

**Proposition 4.1** (Geoffrion, 1974)**.** $z_{LD} = \min\{c^\top x : Ax \geq b,\; x \in \operatorname{conv}(X')\}$. *Hence* $z_{LP} \leq z_{LD}$*, with equality if* $\operatorname{conv}(X') = \{x \geq 0 : Dx \geq d\}$ *(the* integrality property*).*

**Dantzig–Wolfe reformulation.** If $X'$ is bounded with points $\{x^r\}_{r \in \Omega}$, write $x = \sum_r x^r \lambda_r$, $\sum_r \lambda_r = 1$, $\lambda \geq 0$ (convexification). The master LP $\min\{\sum_r (c^\top x^r)\lambda_r : \sum_r (Ax^r)\lambda_r \geq b,\; \sum_r \lambda_r = 1,\; \lambda \geq 0\}$ has value $z_{LD}$ by Proposition 4.1. In routing, $X'$ splits into $K$ identical blocks (one per vehicle), the blocks are aggregated, and the convexity constraints become a single fleet-size constraint $\sum_r \lambda_r \leq K$ (Vanderbeck, 2000; Desrosiers and Lübbecke, 2011). Integrality is imposed on $x = \sum_r x^r \lambda_r$ (*convexification*) or on $\lambda$ (*discretisation*); for 0-1 problems with identical blocks the two lead to the same set-partitioning formulation (Vanderbeck, 2000).

**Lagrangian relaxation versus column generation.** Both compute $z_{LD}$. Subgradient methods are cheap per iteration but converge slowly and give no primal information; column generation gives primal solutions in $\lambda$ that drive branching and cut separation, and Lagrangian bounds are available at every iteration (Proposition 5.1). Barnhart et al. (1998) and Lübbecke and Desrosiers (2005) discuss the relation. Lagrangian relaxation remains useful in BPC as a cheap pre-bound (e.g. dualising capacity and duration in a single-tour relaxation, as in the CPTP-RCTW manuscript) and as a dual-ascent warm start: Baldacci et al. (2011b) compute their bounds with dual ascent procedures before column generation.

**State-space relaxation.** Christofides et al. (1981b) proposed mapping the state space of a dynamic program onto a smaller space so that the relaxed DP gives a bound. Relaxing the visited set of the elementary shortest path to its cardinality, or to its total demand, yields the $q$-paths/$q$-routes of Christofides et al. (1981a): walks whose load is tracked but which may revisit customers. The ng-route and DSSR relaxations of Section 6 are partial state-space relaxations that restore elementarity selectively.

**Set partitioning versus set covering.** With $a_{ir} \in \{0,1\}$ indicating whether route $r$ visits customer $i$, the master constraints $\sum_r a_{ir}\lambda_r = 1$ can be relaxed to $\geq 1$ when costs satisfy the triangle inequality, because a covering solution can be shortcut into a partitioning one of no greater cost. Covering rows have non-negative duals, which stabilises column generation and permits non-elementary columns in relaxed pricing (Barnhart et al., 1998; Desrosiers and Lübbecke, 2011). With profits, a customer's prize is counted once per visit in a route column, so packing rows ($\leq 1$) are the natural choice for optional customers and partitioning rows for mandatory ones (Section 12).

**Elementary versus relaxed bounds.** Let $z_q$, $z_{ng}$ and $z_{el}$ be master LP values with $q$-route, ng-route and elementary columns. Since the column sets are nested, $z_q \leq z_{ng} \leq z_{el}$ (for minimisation). Baldacci et al. (2011b) introduced ng-routes precisely as a trade-off between the bound quality of elementary routes and the pricing cost of $q$-routes, and Contardo et al. (2015) studied strategies to reach the elementary bound in the VRPTW.

# 5 Column generation in depth

**Master and restricted master.** We use the routing master as a running example. Let $\Omega$ be the set of feasible routes, $c_r$ the cost of route $r$, and $a_{ir}$ the number of visits of $r$ to customer $i \in \mathcal{N}$. The *master problem* (MP) and its dual (DMP) are

$$\begin{aligned} z_{MP} = \min\ & \sum_{r\in\Omega} c_r\lambda_r && z_{MP} = \max\ \sum_{i\in\mathcal{N}} \pi_i - K\pi_0 \\ \text{s.t.}\ & \sum_{r\in\Omega} a_{ir}\lambda_r = 1 \quad (\pi_i),\ i\in\mathcal{N} && \text{s.t.}\ \sum_{i\in\mathcal{N}} a_{ir}\pi_i - \pi_0 \leq c_r,\ r\in\Omega \\ & \sum_{r\in\Omega} \lambda_r \leq K \quad (\pi_0 \geq 0) && \pi \text{ free},\ \pi_0 \geq 0 \\ & \lambda \geq 0. \end{aligned} \tag{3}$$

The *restricted master problem* (RMP) uses a subset $\Omega' \subseteq \Omega$. Its optimal duals $(\pi, \pi_0)$ are feasible for DMP iff no column has negative *reduced cost*

$$\bar{c}_r = c_r - \sum_{i\in\mathcal{N}} a_{ir}\pi_i + \pi_0. \tag{4}$$

The *pricing problem* is $\bar{c}^* = \min_{r\in\Omega} \bar{c}_r$. If $c_r$ is a sum of arc costs, $\bar{c}_r$ is the cost of $r$ on arcs with modified costs $\tilde{c}_{ij} = c_{ij} - \pi_j$ (with $\pi$ of the depot set to zero) plus $\pi_0$, and finding $\bar{c}^*$ is an ESPPRC on $(V, A)$ with negative-cost cycles (Feillet et al., 2004; Irnich and Desaulniers, 2005).

**Generic loop.** Column generation alternates between solving the RMP and pricing, adding columns with $\bar{c}_r < 0$, until $\bar{c}^* \geq 0$; then $z_{RMP} = z_{MP}$. Each iteration is one step of Kelley's cutting-plane method on the dual (Kelley, 1960), which explains both its strengths and its instability.

**The Lagrangian bound.**

**Proposition 5.1** (Lagrangian bound)**.** *Let $\kappa$ be an upper bound on $\sum_r \lambda_r$ in any optimal MP solution (e.g. $\kappa = K$). At any RMP iteration with duals $(\pi, \pi_0)$ and pricing value $\bar{c}^*$,*

$$z_{RMP} + \kappa \min\{0, \bar{c}^*\} \leq z_{MP} \leq z_{RMP}.$$

*Proof.* For an optimal MP solution $\lambda^*$, $z_{MP} = \sum_r c_r\lambda^*_r = \sum_r(\bar{c}_r + \sum_i a_{ir}\pi_i - \pi_0)\lambda^*_r \geq \bar{c}^* \sum_r \lambda^*_r + \sum_i \pi_i - \pi_0 \sum_r \lambda^*_r$. Write $\sum_r \lambda^*_r = \kappa' \leq \kappa$. Then $\sum_i \pi_i - \pi_0\kappa' \geq \sum_i \pi_i - K\pi_0 = z_{RMP}$ when $\kappa' \leq K$ and $\pi_0 \geq 0$, and $\bar{c}^*\kappa' \geq \kappa \min\{0, \bar{c}^*\}$. □

The bound is the Lagrangian function evaluated at the current duals (Lübbecke and Desrosiers, 2005). It permits *early termination*: a node can be pruned as soon as $z_{RMP} + \kappa \min\{0, \bar{c}^*\} \geq \bar{z}$, without waiting for convergence. With integer costs, column generation may also stop when $\lceil z_{RMP} + \kappa \bar{c}^* \rceil = \lceil z_{RMP} \rceil$.

**Tailing-off and dual oscillation.** Column generation typically makes rapid initial progress and then a long tail of small improvements (*tailing-off*); the duals oscillate wildly between iterations and the RMP is highly degenerate (Lübbecke and Desrosiers, 2005; Ben Amor et al., 2009). Remedies are stabilisation, early branching when the Lagrangian bound stops improving, and good initial columns.

**Stabilisation by smoothing.** Wentges (1997) proposed pricing at a convex combination of the current duals and a *stability centre* $\hat{\pi}$ (the duals that gave the best Lagrangian bound so far):

$$\tilde{\pi} = \alpha \hat{\pi} + (1 - \alpha)\pi^{RMP}, \qquad \alpha \in [0, 1). \tag{5}$$

If pricing at $\tilde{\pi}$ finds no column with negative reduced cost with respect to $\pi^{RMP}$ (a *mis-price*), $\alpha$ is reduced or the RMP duals are used; finite convergence is preserved. Pessoa et al. (2018a) automated the choice of $\alpha$ by a sub-gradient-based direction and combined smoothing with penalty methods; VRPSolver uses this automatic smoothing (Pessoa et al., 2020).

**Box and penalty stabilisation.** The *boxstep* method of Marsten et al. (1975) restricts duals to a box around $\hat{\pi}$. du Merle et al. (1999) generalised it by adding to the primal RMP surplus and slack variables $y^-, y^+$ with costs $\delta^{\mp}$ and bounds $\epsilon^{\mp}$, which corresponds to a piecewise-linear penalty on deviations of $\pi$ from $[\delta^-, \delta^+]$ in the dual. Ben Amor et al. (2009) analysed the choice of explicit stabilising terms.

**In-out separation.** Ben-Ameur and Neto (2007) accelerate cutting-plane and column generation methods by separating at a point on the segment between an interior ("in") point and the current LP optimum ("out"), which is the cutting-plane view of smoothing. Interior-point-based alternatives produce well-centred duals directly; see the primal–dual column generation of Gondzio et al. (2013).

**Dual-optimal inequalities.** Ben Amor et al. (2006) add to the dual inequalities that are satisfied by at least one optimal dual solution, cutting off useless dual regions; in the primal these are extra columns. Gschwind and Irnich (2016) revisited such dual inequalities. For cutting stock, the inequalities $\pi_i \leq \pi_j$ when item $i$ is smaller than $j$ are the classical example (Ben Amor et al., 2006).

**Dynamic constraint aggregation.** Elhallaoui et al. (2005) aggregate set-partitioning rows of customers that are always covered together in the current columns, reducing degeneracy; rows are disaggregated when needed.

**Initialisation and Farkas pricing.** At the root, and especially after branching, the RMP may be infeasible: no subset of current columns covers every partitioning row, or branching has fixed out columns. Two remedies exist.

- *Big-M artificials.* Add a column per row with large cost $M$; the RMP is always feasible, and artificials are driven out by pricing. Too small an $M$ may give wrong bounds while artificials remain positive; too large an $M$ produces extreme duals and poor columns.
- *Farkas pricing.* Use the certificate of infeasibility of the RMP to price.

**Proposition 5.2** (Farkas pricing)**.** *Let the RMP be* $\{\lambda \geq 0 : A_{\Omega'}\lambda = b\}$ *(inequalities can be handled via slacks). By Farkas' lemma it is infeasible iff there is* $y$ *with* $y^\top A_{\Omega'} \leq 0$ *and* $y^\top b > 0$*. The full MP is feasible only if some column* $a_r$*,* $r \in \Omega \setminus \Omega'$*, satisfies* $y^\top a_r > 0$*. Hence the* Farkas pricing problem *is*

$$\min_{r\in\Omega} \; -y^\top a_r, \tag{6}$$

*i.e. the ordinary pricing problem with all original costs set to zero and the dual vector replaced by* $y$*. If its optimum is* $\geq 0$*, the node is infeasible and can be pruned.*

*Proof.* If $y^\top a_r \leq 0$ for all $r \in \Omega$ then $y$ certifies infeasibility of the full MP. Otherwise any column with $y^\top a_r > 0$ violates the certificate and must be added for the certificate to be invalidated. □

Farkas pricing needs no parameter and returns columns that attack infeasibility directly. It is offered as a separate callback by SCIP, whose documentation notes that it equals regular pricing with a zero objective (The SCIP Optimization Suite developers, 2026; Achterberg, 2009), and it is the natural Phase I of a two-phase column generation. For the CPTP-RCTW, infeasibility arises only from mandatory rows (packing rows are satisfied by $\lambda = 0$), so $y$ is supported on mandatory customers and the fleet row (Section 12).

**Column management.** Practical codes add several negative columns per pricing call (often the best few labels), keep a column pool from which columns are re-priced before calling the exact pricer, and remove columns whose reduced cost has stayed large for many iterations (column clean-up, also used in VRPSolver according to Pessoa et al. (2020)). Selecting which generated columns to add is itself a decision; Morabit et al. (2021) learned a column-selection policy and report up to 30% reductions in computing time on their test sets.

**Heuristic versus exact pricing.** Exactness is needed only to certify $\bar{c}^* \geq 0$ (or to compute a valid Lagrangian bound). A hierarchy of heuristic pricers is therefore used first: re-pricing the pool, labeling with restricted successor lists or relaxed dominance, metaheuristics such as the tabu search of Desaulniers et al. (2008), and finally exact labeling. Any column with negative reduced cost may be added; only the final, exact call determines the bound.

# 6 Pricing: the ESPPRC and its relaxations

**Resource-constrained shortest paths.** A *label* $\ell$ at vertex $v(\ell)$ represents a partial path from the source and stores its reduced cost $\bar{c}(\ell)$ and a vector of resource consumptions $T(\ell)$ (e.g. load $q$, time $t$, other resources), plus the set of visited customers for elementarity. Extending $\ell$ along arc $(i, j)$ applies *resource extension functions* (REFs) (Irnich and Desaulniers, 2005). For the VRPTW,

$$q(\ell') = q(\ell) + q_j \leq Q, \qquad t(\ell') = \max\{a_j,\ t(\ell) + s_i + t_{ij}\} \leq b_j, \qquad \bar{c}(\ell') = \bar{c}(\ell) + \tilde{c}_{ij}. \tag{7}$$

Desrochers et al. (1992) solved the non-elementary SPPTW by pseudo-polynomial dynamic programming; Feillet et al. (2004) gave the exact elementary version, introducing *unreachable* customers as a strengthened elementarity resource. Ioachim et al. (1998) treated linear node costs in time.

**Dominance.** $\ell_1$ dominates $\ell_2$ (same vertex) if every feasible completion of $\ell_2$ is feasible for $\ell_1$ at no higher cost. With non-decreasing REFs, a sufficient condition is

$$\bar{c}(\ell_1) \leq \bar{c}(\ell_2), \quad T(\ell_1) \leq T(\ell_2), \quad U(\ell_1) \subseteq U(\ell_2), \tag{8}$$

where $U$ is the visited (or unreachable) set in the elementary case (Irnich and Desaulniers, 2005). Label-setting algorithms process labels in increasing order of a monotone resource (e.g. time); label-correcting algorithms process them in any order and may revisit.

**Algorithm 2** Monodirectional labeling for the ESPPRC (forward, label-correcting)

**Require:** graph $(V, A)$, reduced costs $\tilde{c}$, REFs, ng-sets $N_i$ (or $N_i = V$ for elementary)
$\mathcal{U} \leftarrow \{\ell_0\}$ with $v(\ell_0) = 0$, $\bar{c} = 0$, $T = 0$, $\Pi = \{0\}$; $\mathcal{P}_v \leftarrow \emptyset$ for all $v$
**while** $\mathcal{U} \neq \emptyset$ **do**
  remove a label $\ell$ from $\mathcal{U}$ (e.g. smallest time or cost)
  **for** $(v(\ell), j) \in A$ with $j \notin \Pi(\ell)$ **do** ▷ ng-feasibility
    $\ell' \leftarrow \text{extend}(\ell, j)$ by the REFs; **if** infeasible **then continue**
    $\Pi(\ell') \leftarrow (\Pi(\ell) \cap N_j) \cup \{j\}$
    **if** $\bar{c}(\ell') + \text{LB}_j(T(\ell')) \geq \theta$ **then continue** ▷ completion bound
    **if** no $\ell'' \in \mathcal{P}_j$ dominates $\ell'$ **then**
      remove from $\mathcal{P}_j$ and $\mathcal{U}$ all labels dominated by $\ell'$; add $\ell'$ to $\mathcal{P}_j$ and $\mathcal{U}$
**return** labels in $\mathcal{P}_{n+1}$ with $\bar{c} < 0$

**Bidirectional labeling.** Righini and Salani (2006) extend labels forward and backward only up to a half-way point of a monotone resource and then join forward and backward labels, reducing the exponential growth of label counts with path length. Tilk et al. (2017) choose the half-way point dynamically from the numbers of labels processed in each direction. Gschwind et al. (2018) studied bidirectional labeling for pickup-and-delivery pricing. Joining requires a feasibility and cost test on pairs; bounds on the backward cost prune forward labels that cannot be joined profitably.

**Completion bounds.** A lower bound $\text{LB}_j(T)$ on the reduced cost of any feasible completion from $j$ with consumption $T$ allows pruning of $\ell$ if $\bar{c}(\ell) + \text{LB}_{v(\ell)}(T(\ell)) \geq \theta$, where $\theta$ is 0 or the threshold for useful columns. Such bounds come from relaxed backward DPs: $q$-routes or ng-routes computed backward on (vertex, discretised resource) states (Baldacci et al., 2011b; Contardo et al., 2015). Bounds that ignore resources (pure shortest paths) are valid but weak.

**Decremental state-space relaxation (DSSR).** Boland et al. (2006) and Righini and Salani (2008) start with elementarity imposed only on a small *critical set* $\Theta$ of customers; if the optimal path contains a cycle, the repeated customers are added to $\Theta$ and pricing is repeated. Righini and Salani (2009) applied DSSR to the orienteering problem with time windows.

**$k$-cycle elimination.** Forbidding cycles of length at most $k$ in non-elementary pricing was used early in VRPTW column generation (2-cycle elimination) and generalised to $k \geq 3$ by Irnich and Villeneuve (2006). It strengthens $q$-route bounds but is dominated in practice by ng-routes.

**ng-route relaxation.** Baldacci et al. (2011b) associate with each customer $i$ a *neighbourhood* $N_i \subseteq \mathcal{N}$, $i \in N_i$, typically the $\Delta$ nearest customers.

**Definition 6.1** (ng-path)**.** For a path $P = (0, v_1, \ldots, v_k)$ define the memory $\Pi(P)$ recursively by $\Pi((0)) = \emptyset$ and $\Pi(P \oplus j) = (\Pi(P) \cap N_j) \cup \{j\}$. $P$ is an *ng-path* if $v_{h+1} \notin \Pi((0, \ldots, v_h))$ for every $h$; an ng-route is an ng-path from depot to depot.

A customer $j$ may be revisited only if, since its last visit, the path has passed through a customer whose neighbourhood does not contain $j$: the path "forgets" $j$. Elementary routes are ng-routes; with $N_i = \mathcal{N}$ ng-routes are exactly the elementary routes, and with $N_i = \{i\}$ only 1-cycles are forbidden. In labeling, $\Pi(\ell)$ replaces the visited set both in extension and in dominance: $\ell_1$ dominates $\ell_2$ if (8) holds with $U = \Pi$. Using the visited set for extension and $\Pi$ for dominance is invalid for elementary pricing (a dominated label could be the only one allowed to visit a customer later) and gives no relaxation benefit—a point we return to in Section 12.

*Dynamic* ng-sets are enlarged where the LP solution contains cycles, combining ng with DSSR (Martinelli et al., 2014; Contardo et al., 2015); arc-memory variants are used for cuts (Pecin et al., 2017a).

**Bucket graphs.** Sadykov et al. (2021) group labels by vertex and by intervals of the main resource(s) into *buckets*, organised in a bucket graph on which labeling becomes a label-correcting DP. This reduces the number of dominance checks, exploits symmetry between forward and backward directions, and supports bucket-arc elimination; the benefits are largest for instances with large capacities or wide time windows.

**Reduced-cost arc fixing.** Given a gap $\Delta = \bar{z} - z_{LB}$ and a lower bound $\bar{c}_{ij}$ on the reduced cost of any route using arc $(i, j)$, the arc can be removed if $\bar{c}_{ij} > \Delta$ (for minimisation, after convergence of column generation): no route through it can be part of an improving solution. $\bar{c}_{ij}$ is computed by joining forward and backward completion bounds through $(i, j)$; Irnich et al. (2010) formalised path reduced costs for this purpose, Sadykov et al. (2021) eliminate bucket arcs, and Desaulniers et al. (2020) fix two-arc sequences.

**Route enumeration.** When $\Delta$ becomes small, all elementary routes with reduced cost at most $\Delta$ may be enumerated into a pool; pricing then becomes inspection of the pool and the master can be passed to a MIP solver once the pool is small enough (Baldacci et al., 2008; Contardo and Martinelli, 2014; Pecin et al., 2017b). Enumeration interacts with cuts: non-robust cuts are handled exactly by the pool.

**Other exact ESPPRC algorithms.** Lozano et al. (2016) proposed the pulse algorithm, a depth-first implicit enumeration with bounding and rollback pruning. Martinelli et al. (2014) combine elementary and restricted non-elementary pricing.

# 7 Cutting planes in a column-generation master

**Where cuts live.** Let $x_{ij} = \sum_r b^r_{ij} \lambda_r$, where $b^r_{ij}$ counts traversals of arc $(i, j)$ by route $r$. A cut $\sum_{(i,j)} \alpha_{ij} x_{ij} \geq \beta$ on the compact arc variables becomes the master row $\sum_r (\sum_{ij} \alpha_{ij} b^r_{ij}) \lambda_r \geq \beta$ with dual $\sigma \geq 0$, and it changes arc reduced costs to $\tilde{c}_{ij} - \alpha_{ij}\sigma$. Such cuts are *robust*: the pricing problem keeps its structure (Fukasawa et al., 2006; Poggi and Uchoa, 2014). A cut whose coefficient for column $r$ is not a linear function of $b^r$ (e.g. involves rounding of route-dependent quantities) is *non-robust*; its dual cannot be pushed to arcs, and pricing must track additional state (Jepsen et al., 2008; Desaulniers et al., 2011). Non-robust cuts are often much stronger per cut but can make labeling exponentially harder, so their number and support must be controlled.

**Rounded capacity cuts (RCC).** For $S \subseteq \mathcal{N}$ with demand $q(S)$,

$$x(\delta(S)) \geq 2 \lceil q(S)/Q \rceil . \tag{9}$$

They are robust and are separated heuristically, e.g. by the CVRPSEP package of Lysgaard et al. (2004), which also separates framed capacity, strengthened comb, multistar and hypotour inequalities. Ralphs et al. (2003) studied separation of capacity constraints and branch-and-cut for the CVRP.

**Framed capacity and comb inequalities.** Framed capacity inequalities strengthen RCC using a partition of $S$ into subsets whose bin-packing requirement exceeds the RCC right-hand side; comb inequalities transfer TSP combs to the CVRP. Both are robust; their separation in CVRPSEP is heuristic (Lysgaard et al., 2004).

**Multistar / generalised large multistar (GLM).** Multistar inequalities exploit that edges between $S$ and outside customers carry demand into $S$'s routes (Letchford et al., 2002). The GLM inequality of Gouveia (1995) reads

$$x(\delta(S)) \;\geq\; \frac{2}{Q}\Big(q(S) + \sum_{i\in S}\sum_{j\in\mathcal{N}\setminus S} q_j\, x_{ij}\Big), \tag{10}$$

which is robust, with arc coefficient $1 - 2q_j/Q$ on edges $\{i,j\}$, $i \in S$, $j \notin S \cup \{0\}$, and 1 on depot edges.

**Lifted cover inequalities (LCI) and where they apply.** For a knapsack $\sum_{j\in J} w_j z_j \leq W$ in binaries $z$, a *cover* $C \subseteq J$ has $w(C) > W$; the cover inequality $\sum_{j\in C} z_j \leq |C| - 1$ is valid and can be lifted to $\sum_{j\in C} z_j + \sum_{j\notin C} \alpha_j z_j \leq |C| - 1$ (Balas, 1975; Wolsey, 1975; Padberg, 1975); Gu et al. (1998) discuss computation. In BPC, LCIs are useful only on knapsacks that are *not* convexified by the pricing problem. A route's own capacity constraint is already enforced in $\Omega$, so covers on it are redundant in the master. LCIs apply to knapsacks that couple several columns or that live in the compact variables: arc-capacity knapsacks in multicommodity flow, where Barnhart et al. (2000) used them in a BPC; or, in selective routing, the aggregate knapsack $\sum_i q_i y_i \leq KQ$ (or $\leq Q$ for $K = 1$) on visit variables $y_i = \sum_r a_{ir}\lambda_r$, which is robust since $y$ is linear in $\lambda$.

**Clique inequalities.** Spoorendonk and Desaulniers (2010) derive clique inequalities from conflicts between customers in the set-partitioning master of the VRPTW and handle their duals in pricing.

**Subset-row and rank-1 cuts.** Jepsen et al. (2008) apply Chvátal–Gomory rounding to master rows. For $S \subseteq \mathcal{N}$ and multiplier $1/k$, the *subset-row inequality* is

$$\sum_{r\in\Omega} \Big\lfloor \frac{1}{k} \sum_{i\in S} a_{ir} \Big\rfloor \lambda_r \leq \Big\lfloor \frac{|S|}{k} \Big\rfloor, \tag{11}$$

most commonly with $|S| = 3$, $k = 2$: the total weight of routes visiting at least two customers of $S$ is at most one. More generally, a *rank-1* (Chvátal–Gomory) cut with multipliers $p \in [0,1)^{|\mathcal{N}|}$ reads $\sum_r \lfloor \sum_i p_i a_{ir} \rfloor \lambda_r \leq \lfloor \sum_i p_i \rfloor$ (Petersen et al., 2008; Pecin et al., 2017b). These cuts are non-robust. With dual $\sigma_S \leq 0$ (for a $\leq$ row in a minimisation), each label carries a state $s_S \in \{0, \ldots, k-1\}$ counting visits to $S$ modulo $k$; when the count reaches $k$ the reduced cost increases by $-\sigma_S$ and the state resets. Dominance must account for states: Jepsen et al. (2008) show that $\ell_1$ dominates $\ell_2$ if

$$\bar{c}(\ell_1) - \sum_{S:\, s_S(\ell_1) > s_S(\ell_2)} \sigma_S \leq \bar{c}(\ell_2) \tag{12}$$

together with the other conditions in (8); requiring equal states is valid but much weaker.

**Limited-memory rank-1 cuts.** Pecin et al. (2017b) reduce the pricing impact of rank-1 cuts by giving each cut a *memory set* $M \supseteq S$: a label's state for the cut is reset to zero whenever the path leaves $M$. The resulting cut is weaker than the full-memory one in general but valid, and the memory is chosen just large enough to keep the cut violated by the current LP solution. Pecin et al. (2017a) use arc memory instead of vertex memory for the VRPTW, with inequalities derived from Balas' elementary inequalities, in a BPC with bidirectional labeling, DSSR, completion bounds, variable fixing and route enumeration; they report solving all 56 Solomon instances with 100 customers and 51 of the 60 Gehring–Homberger instances with 200 customers.

**Other non-robust families.** Kohl et al. (1999) introduced 2-path cuts (robust in arc variables but separated with route feasibility checks); Desaulniers et al. (2008) generalised $k$-path inequalities and partial elementarity; Desaulniers et al. (2011) survey cuts for B&P, and Petersen et al. (2008) test Chvátal–Gomory rank-1 cuts on VRPTW masters.

**Separation strategy.** Typical policies: separate robust cuts first, add non-robust cuts only when column generation has converged and the robust loop tails off; limit the number of cuts per round and per customer; filter near-parallel cuts by an orthogonality test; and purge inactive cuts. After adding cuts, column generation must be resumed because the duals change.

# 8 Branching and primal heuristics

**Why not branch on $\lambda$.** Fixing $\lambda_r = 0$ requires the pricer to avoid regenerating $r$, i.e. to solve a $k$-th shortest path problem, and the branch is very unbalanced. Branching is therefore done on compact quantities that the pricer can respect (Barnhart et al., 1998; Vanderbeck, 2000).

**Vehicle-count branching.** If $\sum_r \lambda_r = \kappa^* \notin \mathbb{Z}$, create $\sum_r \lambda_r \leq \lfloor \kappa^* \rfloor$ and $\geq \lceil \kappa^* \rceil$. The rows enter the master; their duals are constants in pricing.

**Arc/edge branching.** If $x_{ij} = \sum_r b_{ij}^r \lambda_r$ is fractional, set $x_{ij} = 0$ (remove the arc) or $x_{ij} = 1$ (remove all other arcs out of $i$ and into $j$). Both are robust. Branching on cut-set values ($x(\delta(S)) = 2$ versus $x(\delta(S)) \geq 4$) is a stronger alternative used by Fukasawa et al. (2006).

**Ryan–Foster branching.** For a set-partitioning master, Ryan and Foster (1981) show that a fractional solution has rows $i, j$ with $0 < \sum_{r:\, a_{ir}=a_{jr}=1} \lambda_r < 1$. Branch on "$i$ and $j$ together" versus "$i$ and $j$ apart". In routing pricing, "together" forces $j$ whenever $i$ is visited and vice versa; "apart" forbids visiting both. Pessoa et al. (2020) generalise this rule within VRPSolver via packing sets.

**Resource and time-window branching.** Gélinas et al. (1995) branch on time windows: split $[a_i, b_i]$ at a value separating the arrival times of fractional routes at $i$. Pessoa et al. (2020) branch on accumulated resource consumption. Such rules act on the pricer's resource bounds and keep the structure.

**Strong branching and pseudo-costs in BPC.** Because node LPs are expensive, BPC codes use multi-phase strong branching: evaluate many candidates by a cheap proxy (pseudo-costs or RMP re-solves without pricing), then a few with heuristic column generation, and finally one or two with full column generation (Pecin et al., 2017b; Pessoa et al., 2020). Pseudo-costs (Benichou et al., 1971) and reliability rules (Achterberg et al., 2005) transfer directly.

**Primal heuristics.** The *restricted master heuristic* solves the integer RMP over the current column pool with a MIP solver. *Diving* fixes columns or compact variables and resumes column generation without backtracking (or with limited backtracking); Sadykov et al. (2019) show the assets of diving in B&P for generalised assignment, cutting stock and vertex colouring. Upper bounds may also come from external metaheuristics, in which case BPC proves optimality.

# 9 The assembled algorithm

**Validity.** BPC is a B&B in which bounds come from Proposition 5.1 or a converged RMP, cuts are globally valid or valid in the subtree, and branching decisions are enforced in pricing.

**Algorithm 3** Processing a node in branch-price-and-cut

**Require:** node $k$ with branching decisions $\mathcal{B}_k$, column pool $\Omega'$, cut pool $\mathcal{C}$, incumbent value $\bar{z}$

optionally compute a cheap Lagrangian bound; **if** $\geq \bar{z}$ **then** prune
remove columns violating $\mathcal{B}_k$; restrict pricing graph accordingly
**repeat** ▷ cut loop
    **repeat** ▷ column generation
        solve RMP; **if** infeasible: Farkas pricing (6); **if** none, prune
        compute stabilised duals $\tilde{\pi}$ (5)
        heuristic pricing → exact pricing on the ng-relaxed ESPPRC with cut states
        update Lagrangian bound $z_{LB}$ (Proposition 5.1); **if** $z_{LB} \geq \bar{z}$, prune
        add columns with $\bar{c} < 0$ **until** no column found **or** tailing-off
    **until** converged
    reduced-cost arc fixing with gap $\bar{z} - z_{LB}$; optionally enumerate routes if gap small
    separate robust then non-robust cuts; add the most violated, orthogonality-filtered
**until** no violated cut **or** bound improvement below tolerance
restricted-master / diving heuristic; update $\bar{z}$
**if** LP solution integral **then** update incumbent; prune
**else** choose branching candidate by multi-phase strong branching; create children

Correctness requires: (i) the exact pricer is exact for the column set implied by the formulation (elementary or ng-relaxed, with all active cut and branching states); (ii) column fixing and arc elimination use valid bounds; (iii) a node is declared converged only after an exact pricing call returns no negative column at the true RMP duals, not the smoothed ones.

**Interaction of components.** Cuts tighten the bound but enlarge the pricing state (non-robust) or change arc costs (robust); arc fixing and enumeration shrink pricing as the gap closes; branching on compact quantities modifies the pricing graph or resources; stabilisation mostly affects the number of iterations, not the final bound. The design problem is to balance bound strength against pricing cost; this is what limited memory, ng-sets and bucket graphs address (Pecin et al., 2017b; Sadykov et al., 2021; Costa et al., 2019).

# 10 Parameters, engineering and software

**Parameters.** Table 1 lists the knobs a BPC implementation exposes. The right-hand column gives the defaults found in the existing CPTP-RCTW code base and manuscript (`wsmart-route`, BPC configuration and `paper.tex`); the literature column is qualitative, because published tuned values depend on the instance class and are not transferable as numbers.

**Numerical safeguards.** Use a tolerance for negative reduced costs (columns with $\bar{c} > -\epsilon$ are not improving); recompute reduced costs of added columns exactly with the RMP duals; in the Lagrangian bound use the exact (non-smoothed) duals and an exact pricing value; when labeling is truncated by time or label limits, do not use its value as a bound.

**Software.** *VRPSolver* is a generic BCP solver for vehicle routing and related problems built on the BaPCod library; it implements ng-paths, limited-memory rank-1 cuts, path enumeration, rounded capacity cuts, automatic stabilisation, resource-based and generalised Ryan–Foster branching, and models bin packing, vector packing and generalised assignment (Pessoa et al., 2020; Sadykov and Vanderbeck, 2021). *BaPCod* is a C++ library that generates the Dantzig–Wolfe reformulation from a compact model and offers restricted-master and diving heuristics, stabilisation and strong branching (Sadykov and Vanderbeck, 2021). *SCIP* provides the B&B,

Table 1: BPC parameters. "Code" = current default in the CPTP-RCTW implementation/manuscript read for this guide.

| Parameter | Role and trade-off | Code default |
|---|---|---|
| ng-neighbourhood size $\Delta$ | Larger: stronger bound, more labels; dynamic enlargement where cycles appear (Baldacci et al., 2011b; Contardo et al., 2015) | 12 |
| DSSR initial critical set | Size of elementarity set before cycles force additions (Righini and Salani, 2008) | 8 |
| Columns per pricing call | More columns: fewer iterations, larger RMP; selection can be learned (Morabit et al., 2021) | 50 |
| Max CG iterations per node | Guard against tailing-off; use Lagrangian bound for early stop | 150 |
| Smoothing $\alpha$ | Wentges factor; automatic tuning avoids manual choice (Wentges, 1997; Pessoa et al., 2018a) | present (Wentges) |
| Cuts per round | Fewer: cheaper LP and pricing; more: faster bound progress | 10 |
| Orthogonality threshold | Filters near-parallel cuts | 0.7 |
| Non-robust cut limits | Per-customer and total limits on SRC/rank-1; memory size (Pecin et al., 2017b) | $\lvert S\rvert = 3$ only |
| Labels / pricing time limits | Abort heuristic pricing; exact call must not be truncated if the bound is used | $10^6$ labels, 30 s |
| Heuristic successor list | Restrict extensions to nearest successors in heuristic pricing | $\min(20, \max(5, n/3))$ |
| Strong-branching candidates | Phases and candidate counts (Achterberg et al., 2005; Pecin et al., 2017b) | visit vars only |
| Node selection | Best-first vs depth-first/diving (Morrison et al., 2016) | not checked |
| Lagrangian iterations | Subgradient steps for pre-pruning (Held et al., 1974) | 40 |
| Enumeration threshold | Gap/route count at which routes are enumerated (Baldacci et al., 2008; Pecin et al., 2017b) | not implemented |
| Optimality gap / time / nodes | Termination | 0.5%, 60 s, 2000 |

cut and pricer plug-in architecture with reduced-cost and Farkas pricing callbacks (Achterberg, 2009; The SCIP Optimization Suite developers, 2026); *GCG* builds a generic Dantzig–Wolfe B&P on top of SCIP (Gamrath and Lübbecke, 2010). CVRPSEP separates robust CVRP inequalities (Lysgaard et al., 2004).

# 11 Applications

**Capacitated VRP.** Pricing is an ESPPRC with a load resource; ng-routes, RCC, rank-1 cuts with limited memory, enumeration and edge branching form the state of the art (Fukasawa et al., 2006; Baldacci et al., 2008; Pecin et al., 2017b; Pessoa et al., 2020); instance sets include those of Uchoa et al. (2017). Robust BCP over extended formulations also served the capacitated minimum spanning tree (Uchoa et al., 2008), and two-commodity flow branch-and-cut gave an alternative exact CVRP approach (Baldacci et al., 2004).

**VRPTW.** Time is the main resource, which makes label-setting in time natural and dominance effective. Milestones are Desrochers et al. (1992), Kohl et al. (1999), Jepsen et al. (2008), Desaulniers et al. (2008), Baldacci et al. (2011b), Contardo et al. (2015) and Pecin et al. (2017a); benchmarks are from Solomon (1987); surveys by Desaulniers et al. (2014) and Baldacci et al. (2012).

**Other routing variants.** Pickup and delivery with time windows: Ropke and Cordeau (2009) (BCP) and Baldacci et al. (2011a), with bidirectional labeling studied by Gschwind et al. (2018). Split delivery: Desaulniers (2010). Heterogeneous fleets: Pessoa et al. (2018b). Multi-depot with route length: Contardo and Martinelli (2014). Multi-period with due dates: Archetti et al. (2015).

**Routing with profits.** Feillet et al. (2005) classify single-vehicle problems with profits: prize-collecting TSP (Balas, 1989), orienteering (Golden et al., 1987), and the profitable tour problem (PTP) (Dell'Amico, 1995). In column generation, prizes become vertex profits in pricing, so the structure is that of the VRP with optional rows. Boussier et al. (2007) gave a B&P for the team orienteering problem and its time-window variant; Keshtkaran et al. (2016) enhanced exact methods for TOP; Archetti et al. (2009) introduced the capacitated TOP and capacitated PTP; Archetti et al. (2013) studied their optimal solution; Jepsen et al. (2014) gave a branch-and-cut for the single-vehicle capacitated PTP; Bérubé et al. (2009) and Fischetti et al. (1997) give generalised subtour elimination machinery for selective problems. Righini and Salani (2009) solved the orienteering problem with time windows with DSSR. Surveys: Vansteenwegen et al. (2011); Gunawan et al. (2016).

**Crew scheduling.** Pricing is a resource-constrained shortest path in a duty or pairing network with labour rules as resources (Desrochers and Soumis, 1989; Desrosiers et al., 1995; Vance et al., 1997). Ryan–Foster branching originated in this setting (Ryan and Foster, 1981).

**Cutting stock and bin packing.** Pricing is a knapsack (Gilmore and Gomory, 1961, 1963). B&P versions are given by Vance et al. (1994) and Vanderbeck (1999); Belov and Scheithauer (2006) added Chvátal–Gomory cuts in a BCP; Delorme et al. (2016) survey models and exact methods. Dual-optimal inequalities are particularly effective here (Ben Amor et al., 2006).

**Generalised assignment.** Each machine's knapsack defines a block; Savelsbergh (1997) gave a B&P, and GAP is among the problems handled by VRPSolver and by diving heuristics (Pessoa et al., 2020; Sadykov et al., 2019).

**Multicommodity flow and network design.** Path variables per commodity, with arc capacity knapsacks cut by LCIs (Ford and Fulkerson, 1958; Barnhart et al., 2000); in-out separation for network design (Ben-Ameur and Neto, 2007).

# 12 Towards a BPC for the CPTP-RCTW: diagnosis and research agenda

**Problem.** Following the existing manuscript, let $\mathcal{N}$ be customers, $\mathcal{M} \subseteq \mathcal{N}$ mandatory and $\mathcal{O} = \mathcal{N} \setminus \mathcal{M}$ optional; customer $i$ has prize $p_i$, demand $q_i$, service time $s_i$ and window $[a_i, b_i]$; at most $K$ vehicles of capacity $Q$; additional cumulative resources $\rho \in \mathcal{R}$ (route duration, length, number of stops) with arc consumption $\tau_{ij}^{\rho}$ and limit $U^{\rho}$. The objective is to maximise collected prize minus travel cost. Special cases include the PTP, CPTP, CTOP, TOP and TOPTW, and the VRPTW when $\mathcal{M} = \mathcal{N}$.

**Master and pricing.** With $\Omega$ the routes feasible for capacity, windows and $\mathcal{R}$, and $\tilde{p}_r = \sum_i p_i a_{ir} - c_r$:

$$\begin{aligned} \max \ & \sum_{r\in\Omega} \tilde{p}_r \lambda_r \\ \text{s.t.} \ & \sum_{r\in\Omega} a_{ir}\lambda_r = 1 \quad (\pi_i \text{ free}), \quad i \in \mathcal{M}, \\ & \sum_{r\in\Omega} a_{ir}\lambda_r \le 1 \quad (\pi_i \ge 0), \quad i \in \mathcal{O}, \\ & \sum_{r\in\Omega} \lambda_r \le K \qquad (\pi_0 \ge 0), \quad \lambda \ge 0. \end{aligned} \tag{13}$$

The reduced cost is $\bar{c}_r = \tilde{p}_r - \sum_i a_{ir}\pi_i - \pi_0$; pricing is $\max_r \bar{c}_r$, an ESPPRC with time windows on arc profits $p_j - \pi_j - c_{ij}$; and the Lagrangian bound is $z_{MP} \le z_{RMP} + K \max\{0, \bar{c}^*\}$ (Proposition 5.1 with signs reversed). Farkas pricing arises only for mandatory rows, since $\lambda = 0$ satisfies all packing rows.

**Diagnosis of the current implementation.** Reading the pricing, master, separation and branching modules of the code base gave the following findings. They are observations from source code and should be confirmed by tests before being reported elsewhere.

1. *No time or extra resources in the label.* The label stores reduced cost, vertex, load, visited set, ng-memory, Ryan–Foster state and SRI state, but no time, duration or other resource; no time-window or service-time fields occur in the pricing, search or BPC modules. The manuscript describes ng-route labeling with time windows, so the code currently solves the capacitated (CPTP) pricing problem, not the CPTP-RCTW one.
2. *Apparent ng inconsistency.* Extension rejects every vertex in the visited set (full elementarity), and additionally vertices in the ng-memory; dominance with ng enabled compares only ng-memories. As discussed in Section 6, a label may then dominate another that has *not* visited a customer it has visited, so the "exact" pricer may miss columns while gaining no relaxation. With extension fully elementary, DSSR has no effect. This should be confirmed by a test comparing pricing values with ng dominance against elementary dominance on small instances.
3. *Limited-memory rank-1 cuts are not added.* The separation routine calls a method for limited-memory rank-1 cuts that the master does not define; the fallback adds subset-row cuts, which reject $|S| \neq 3$. SRI states are tracked without memory, and dominance requires equal states rather than (12).
4. *Weak completion bounds.* Completion bounds are Bellman–Ford style shortest paths that ignore load, time and non-robust cut duals, and they are used to gate the return to the depot rather than to prune labels during extension.
5. *Missing accelerations.* The labeling is forward-only and label-correcting (no bidirectional search, no bucket graph); there is no route enumeration, no pseudo-cost or reliability branching, and no in-out or box stabilisation. Wentges smoothing and a Farkas Phase I are implemented, as are reduced-cost arc fixing, a Lagrangian pre-bound, and a branching hierarchy (customer visit with strong branching, fleet, divergence, Ryan–Foster, arc).

**Research agenda.** Each idea is tagged [E] (established technique to transfer), [M] (moderate extension of published work) or [S] (speculative; no published support known to us).

**A. Correct and complete pricing.**

- [E] *Time, duration and $\mathcal{R}$ resources in labels* with REFs (7) and dominance (8) (Desrochers et al., 1992; Irnich and Desaulniers, 2005). Route duration with waiting requires care: if duration is measured from the actual departure, the departure time can be postponed, and a label must carry both earliest arrival and accumulated duration with a non-trivial REF; the generic REF framework of Irnich and Desaulniers (2005) covers this.

- [E] *Consistent ng-route pricing*: use $\Pi$ in both extension and dominance (Definition 6.1), with dynamic enlargement of ng-sets and DSSR on top (Baldacci et al., 2011b; Martinelli et al., 2014; Contardo et al., 2015). This is a correctness fix before any performance work.
- [E] *Bidirectional labeling with time as the critical resource* and dynamic half-way points (Righini and Salani, 2006; Tilk et al., 2017); then a bucket graph (Sadykov et al., 2021), whose advantage for wide windows and large capacities matches waste collection with long shifts.
- [M] *Profit-aware completion bounds.* Backward ng- or $q$-route DPs over (vertex, load bucket, time bucket) on arc profits $p_j - \pi_j - c_{ij}$ give upper bounds on completion profit. With optional customers, many completions return directly to the depot, so the bound must allow "stop now" and must include the (non-positive for optional rows) effect of packing duals. These bounds serve label pruning, arc fixing (Irnich et al., 2010; Desaulniers et al., 2020) and enumeration.
- [M] *Mandatory-first heuristic pricing.* In Phase I, Farkas duals are supported on mandatory customers only, so heuristic pricers restricted to the subgraph induced by mandatory customers (plus cheap optional insertions) may reach feasibility faster.

**B. Cuts that see selection, windows and resources.** The manuscript notes that its current families ignore time windows, $\mathcal{R}$ and the mandatory/optional distinction.

- [E] *Proper limited-memory rank-1 cuts* with vertex or arc memory and the state-based dominance (12) (Jepsen et al., 2008; Pecin et al., 2017b,a); fixing the silent drop is a prerequisite.
- [M] *Selection-conditioned capacity cuts.* For $S \subseteq \mathcal{N}$ with optional customers, replace $q(S)$ in (9)–(10) by $\sum_{i \in S} q_i y_i$ (fractional capacity form) and add generalised subtour inequalities $x(\delta(S)) \geq 2y_v$, $v \in S$ (Fischetti et al., 1997; Bérubé et al., 2009; Jepsen et al., 2014); rounding is valid only on the mandatory part.
- [M] *Aggregate resource covers.* The knapsacks $\sum_i q_i y_i \leq KQ$ and, under the triangle inequality, $\sum_i s_i y_i +$ (lower bound on travel time) $\leq KU^T$ are not convexified by pricing and admit robust LCIs in $y$ (Gu et al., 1998).
- [S] *Time-window multistars.* Replace demand in (10) by a time-based quantity (service plus minimal travel) bounded by window widths, to bound how many routes must enter a set of customers with tight, overlapping windows. Validity needs proof.
- [S] *Mandatory–fleet rank-1 cuts.* Rank-1 cuts combining mandatory rows and the fleet row, e.g. with multipliers on the fleet row, to attack fractional fleet use; separation could reuse the rank-1 machinery since the fleet row has coefficient one in every column.

**C. Branching and tree search.**

- [E] *Pseudo-cost/reliability branching* on visit variables $y_i$ and arcs (Benichou et al., 1971; Achterberg et al., 2005), replacing full strong branching at every node.
- [E] *Time-window and resource branching* (Gélinas et al., 1995; Pessoa et al., 2020) as a fallback when selection is integral but routes are fractional.
- [E] *Route enumeration* once the profit gap is small (Baldacci et al., 2008; Pecin et al., 2017b); with optional customers the number of near-optimal routes may be large, so the enumeration threshold must be tested.
- [M] *Diving and restricted-master heuristics* (Sadykov et al., 2019) seeded by the existing ALNS/matheuristics in the code base for strong initial incumbents.

**D. Stabilisation and learning.**

- [E] *Automatic smoothing* (Pessoa et al., 2018a) replacing a fixed $\alpha$; in-out separation for cuts (Ben-Ameur and Neto, 2007).
- [S] *Dual-optimal inequalities for optional rows.* Bounds such as $\pi_i \leq p_i +$ (saving) for optional $i$ could be derived from the fact that dropping $i$ from a route never increases cost under the triangle inequality (Ben Amor et al., 2006; Gschwind and Irnich, 2016); the exact form and validity need proof.

- [M] *Learned column selection and branching* (Morabit et al., 2021; Khalil et al., 2016; Bengio et al., 2021); the code base already contains an ML-guided BPC variant, which should be evaluated against the exact baseline with identical time limits and with correctness preserved (learning only in heuristic decisions).

**E. Problem-level extensions.**

- [S] *Rolling-horizon BPC for smart waste collection* (Ramos et al., 2018; Archetti et al., 2015): reuse column and cut pools across days, with prizes derived from fill-level forecasts; stochastic fill levels could enter via scenario-dependent prizes.

**Suggested order of work.** (1) Fix ng consistency and add time/resource labels; (2) implement Jepsen dominance and real limited-memory rank-1 cuts; (3) profit-aware completion bounds and arc fixing; (4) bidirectional/bucket labeling; (5) new cuts for selection–window interaction; (6) enumeration and branching refinements. Steps (1)–(2) affect correctness and should come with unit tests that compare pricing values against brute-force enumeration on small instances.

# 13 Conclusion

BPC is B&B with an extended-formulation bound computed by column generation and strengthened by cuts. Its power in routing comes from convexifying the hardest route constraints in the pricing problem, from relaxations that keep pricing tractable (ng-routes, limited memory), and from accelerations that exploit a small gap (arc fixing, enumeration). For the CPTP-RCTW, the main gaps between the existing code and this state of the art are correctness of pricing (time resources and ng consistency) and the absence of cuts that use selection, windows and resources jointly; these define a concrete research programme.

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